Nintendo Wii System Menu — Pixel-Faithful Recreation Research

Measurements from supplied reference images + sourced documentation · compiled 2026-08-28

M = my measurement from the reference image (ground truth for this build) · **S** = sourced fact · *est.* = inference / undocumented.

0. Reference image measurements M

`wii-home.png` is **637 × 358 px**, but the bottom **11 px is an OS taskbar**, not Wii UI. **The Wii screen area is 637 × 347 px (aspect 1.836)**. All percentages below are of that 637 × 347 box.

Channel grid — 4 columns × 3 rows

Property	px	% of screen	Notes
Left margin (grid start)	50	7.85 % W	grid spans x 50 → 584
Right margin	53	8.32 % W	grid right edge = 91.7 % W
Top margin (grid start)	30	8.65 % H	
Grid bottom edge	252	72.6 % H	= 27.4 % up from bottom
Tile size	130.5 × 71.3	20.49 % W × 20.55 % H	
Column pitch	134.7	21.14 % W	starts x = 50, 185, 319, 454
Row pitch	75.5	21.76 % H	starts y = 30, 105, 181
Horizontal gap	4	0.63 % W	**3.1 % of tile width**
Vertical gap	4.5	1.30 % H	**6.3 % of tile height** (gaps are equal in px)
Corner radius	≈10	1.57 % W / 2.88 % H	**7.7 % of tile W, 14 % of tile H**. Fitted to a true circular arc.

Tile aspect ratio: 1.83 : 1 measured. This is *not* arbitrary — Nintendo's spec **S** gives the channel icon as **128 × 96 (4:3 mode)** and **170 × 96 (16:9 mode)** = 1.771. My 1.830 ÷ the screenshot's own 1.836/1.778 stretch = **1.772. Exact match.** Use `aspect-ratio: 170 / 96`.

Bottom bar & the blue wave

The bar's top edge is a **single broad central valley**, symmetric about screen centre — *not* a multi-peak sine wave. Flat shoulders left and right (which carry the Wii and envelope buttons), one smooth dip in the middle.

Property	px (y from top)	% from bottom
Bar top edge at the shoulders	259	25.4 % H
Bar top edge at centre (valley floor)	298	14.1 % H
Wave amplitude	39	11.2 % of screen height

Horizontal breakdown (% of screen width): flat **0** → **18.5 %**, descending ramp **18.5** → **32.2 %**, flat valley **32.2** → **67.8 %**, ascending ramp **67.8** → **82.1 %**, flat **82.1** → **100 %**. Each ramp is ~13.7 % of screen width. I fitted the ramp against smoothstep ($3t^2–2t^3$): measured 0.179 / 0.487 / 0.795 vs. ideal 0.163 / 0.475 / 0.795 — **it is a smoothstep / cosine ease, i.e. an SVG cubic or `cubic-bezier(.42,0,.58,1)` profile**.

The stroke itself is 3 px at this scale (~0.9 % of H): a 1 px pale halo, a 1 px bright cyan highlight, then a 1 px saturated teal core.

Buttons, clock, cursor

Element	px	% of screen
Wii button (outer ring)	59 W × 61 H	**17.6 % of screen height** (9.2 % of W)
Wii button centre	x 57, y 302	8.9 % from left, 13.0 % up from bottom
Envelope button	58 W × 61 H	17.6 % of H; centre x = 90.8 % (9.2 % from right)
SD-card icon	26 × 35, x 117–143	18.4 – 22.4 % W, y 85.6 – 95.7 % H

Clock digits "05:22"	97 W × 29 H, y 261–289	digit height **8.4 % of H**; font-size ≈ **11.5 % of H**; block width 15.2 % W
Date "Tue 15/7"	96 W × 18 H, y 313–330	font-size ≈ **5.2–5.5 % of H**
Cursor glyph (`wii-cursor.png`)	92 × 128 in a 250² canvas	bbox x 99–191, y 109–237

> **Critical layout detail:** the clock digits sit *above* the wave line, floating on the striped grey background inside the valley. The date sits *below* the line, inside the bar. Both are centre-aligned to the screen (±1 px).

Sampled colours M

Element	Hex	Notes
Background stripes	#F9F9F9 ↔ #E0E0E0	Horizontal 1 px hairlines, 10 px period (2.9 % of H) with *two* bright lines per period at 3 px + 7 px spacing. Not a flat grey.
Bar fill (vertical gradient)	#CBCCD1 → #D8D9DE → #CED1D8	top → ~20 % down → bottom. Cool grey, *not* white; brighten ~8 % if the render looks muddy.
Wave core / highlight / halo	#41A0BB · #79D3ED · #BAE1F0	Highlight sits above the core.
Wii / envelope button ring	#5AA2B8	~1.5 px stroke, same hue family as the wave
Button fill	#E2E2E2 → #CDCEC9	top→bottom, plus a soft concentric outer bezel ring
"Wii" wordmark	#8E8E8E	anti-aliased; true value likely ≈ #7A7A7A
Clock digits	#6E6E6E	neutral grey
Date text	#77787C	very slightly cooler than the clock
Tile rim	#F5F6F5	1–2 px only
Cursor	#008CFF number · #BADFFF→#CAE7FF→#EFF7FF body · #000 outline	drop shadow ≈ rgba(0,0,0,.20)
Boot screen	#000000 bg · #FFED26 triangle · #FFF text	content occupies rows 12 – 87 % of frame height

1. Channel tile visuals

- **M** In this reference there is **no thick silver frame, no gloss overlay, and no drop shadow**. I sampled the flat Discord tile across its full width and height: constant `#5166F7`, zero gradient. The gap under a tile is bit-identical to the background stripes.
- **M** The only edge treatment is a **1–2 px near-white rim** (#F5F6F5). Adjacent rims merge, so the vertical gutters read as a bright 4 px line while the horizontal gutters show background stripes through.
- **S** On real hardware the icon is a live NW4R layout (`icon.brlyt` + `icon.brlan`) drawn into a fixed "Channel aperture"; the art is clipped to the rounded aperture, and animation is *mandatory* — Nintendo's spec explicitly prohibits still-image icons. The subtle bevel people remember comes from the aperture mask, not from the developer's art.
- Practical build: rounded-rect clip + `box-shadow: inset 0 0 0 1.5px rgba(255,255,255,.9)`. Add shadow/gloss only on the hover state.

2. Hover animation

Not formally documented anywhere I could find — but the audio literature corroborates the behaviour precisely: the hover sound is described as "*a low frequency sound that rises in pitch to correspond with the menu item **inflating***" **S**. So the tile genuinely inflates rather than merely brightening.

- **Scale ≈ 1.06–1.10** (*est.*), tile grows about its own centre and overlaps its neighbours — the surrounding grid does **not** reflow or push apart.
- The tile **lifts**: it gains a soft drop shadow and z-index above its neighbours.
- There **is** an overshoot — a single soft rebound, not a repeating jiggle. Character is a spring settling in one bounce. *est.* ~180–220 ms total, e.g. `cubic-bezier(.34,1.56,.64,1)`.
- Un-hover is faster and non-springy, ~120–150 ms ease-out.
- The 4:3 vs 16:9 aperture is fixed, so a hovered tile scales its *whole frame*, art included — no separate parallax layer.

3. Channel launch animation

This is documented in Nintendo's *Icon and Banner Specifications* (RVL-06-0166-001-L, v1.0.0, Feb 2008) **S**:

- Clicking an icon plays a **zoom-in**: the tile expands from its grid position to fill the whole screen, cross-dissolving into the channel's **banner** (`banner.brlyt` / `.brlan`), a separate, higher-resolution full-screen layout. The spec's exact wording: "*Sound starts to play when the banner zoom-in completes.*" There is no white flash — it's a scale + crossfade.
- The destination is the **"Channel Preview screen"** (Nintendo's official name, per the Wii Operations Manual **S**): full-bleed animated banner, channel title top-centre, Nintendo logo top-right, blue triangular page arrows at mid-left/mid-right, and two rounded pill buttons bottom-centre — **"Wii Menu"** (left) and **"Start"** (right) — over a black letterbox frame. Your `wii-mii-for-team-page.jpeg` is exactly this screen.
- Banner animation uses two tags: **Start** plays once, then **Loop** repeats indefinitely.
- Reverse: selecting "Wii Menu" **fades the banner and its sound out together** and zooms back down to the tile. Selecting "Start" cuts the banner sound after **2 seconds** (spec-stated) and fades to the game.
- Render target **S**: framebuffer **608 × 456**, VI **670 × 456**, **59.94 Hz NTSC**. PAL50 runs the same animations at **1.2×** speed.

4. Bottom bar behaviours

- **S** Contents, left → right: **Wii button** (opens Wii Options = Wii Settings + Data Management), **SD Card Menu** icon (*rendered grey when no card is inserted*), **current time** above and **current date** below in the centre, **Wii Message Board** envelope at right.
- **S** **The envelope blinks when a message has been received** — that is the bar's one documented idle animation. Everything else in the bar is static.
- Wii button press *(est.)*: ring brightens and the fill inverts to a pressed/darker gradient, ~100 ms, then the Wii Options screen wipes in over the menu. The Options screen is a light blue-grey panel with two large rounded buttons ("Wii Settings", "Data Management") plus a "Wii Menu" back button at bottom-left.
- Envelope press: the board slides **up from the bottom** like a paper sheet, letterbox-framed.
- **S** Page arrows are **blue**; capacity is **48 channels** = 4 pages of the 4 × 3 grid.

5. The blue wave

Static geometry. In the base system theme the curve does **not** undulate or animate — it is one fixed silhouette (I found no source describing motion, and no theme asset implies a looping curve). What reads as "alive" is the striped background plus the animated channel icons above it.

Visual character, measured: a hairline stroke with a vertical 3-stop build (halo → cyan highlight → teal core, top to bottom), no horizontal gradient across the width; opacity is fully opaque at the core and the halo is the antialiased spill. **One valley, two shoulders — zero repeating peaks.**

6. Sounds

Best analytical source is the "Beyond the Beep" teardown of the Wii's UI audio **S**; direct quotes below.

Sound	Documented character

Startup chime	"a short piece of airy and slightly mystical music that's **barely over one second long**", "gentle synth sounds with a **soft attack**" and a "**heavy amount of delay**". Harmony: an **inverted G major (B–G–D) → inverted Gadd2 (B–D–G–A)**.
Menu ambient loop	"mellow, slowly modulating chords", endlessly looping, deliberately written not to irritate over long sessions. Kazumi Totaka; **C major, ~115 BPM, 4/4** (community transcription — treat the BPM as approximate). Instrumentation: soft mallet/celesta-like chimes over warm pad synths, with a light shaker/pizzicato bass pulse. Loop length is on the order of 30–40 s; note the icon spec's own example loop is frames 0→2400 = **40.0 s at 60 fps**.
Channel hover	"a light and slightly **metallic click**" with "the smallest weight to it"; plus the inflate "**bloop**: a low frequency sound that rises in pitch to correspond with the menu item inflating."
Channel click / launch	"quick **zap**" with "fast attack and sharp texture", followed by an "**energetic, rising woosh** sound that brings you to the landing page."
Returning to menu	"**falling woosh** sound" built from a "high to low **filter sweep**" — the mirror of the launch woosh.
Page scroll	"a very textural quick **series of three clicks**, which feel a bit **wooden and percussive**."
Back / Wii-button-ish	System back cue: "short and mellow beep, played with the note **D3**". The select cue is a longer sustain on **G4** — the two form a perfect 4th/5th.
Cursor movement	No sound. Moving the pointer over empty background is silent; audio fires only on focus (hover) and activation. Confirmed by the Shop Channel's documented UI sound table, which has entries only for hover, decide, cancel, choice-change and error.

Where the rips live: the system menu bank is `IplSound.brsar`. Catalogued at **The Sounds Resource** (sounds.spriters-resource.com, Wii → Wii Menu), the **Internet Archive** ("All Wii Sounds"; "Wii System Soundtrack" FLAC by Kazumi Totaka), **KHInsider** ("Wii System Menu (gamerip)"), **Myinstants**, and GBAtemp threads on Wii Menu sound replacement. Note that BRSAR extraction is notoriously flaky — VGMTrans crashes on export and BrawlBox often emits nothing, so prefer the pre-ripped archives.

7. Fonts

Where	Real font	Free web equivalent
All UI text, channel titles, date, "Wii" wordmark	**Rodin NTLG** (Fontworks / Yutaka Satō, Type Labo, 1997) **S** — also the GameCube, DSi, 3DS and Wii U system font. *(Trivia: Wii menu v0.1 used Futura Pro Bold Italic as a placeholder.)*	**M PLUS Rounded 1c** (Google Fonts, 7 weights) is the standard substitute — humanist, slightly rounded terminals, matching Latin + Japanese coverage. Second choice: **M PLUS 1p** (less rounded) or **Zen Maru Gothic**.
Clock digits — confirmed distinct **M**	Not Rodin. I zoomed the reference 6×: these are true **7-segment LCD glyphs** with chamfered segment ends, visible inter-segment gaps, and a colon of two square dots. Nintendo's own bitmap banner fonts (`wbf1/wbf2.brfna`) are a separate family, so the clock is a dedicated segment face.	**DSEG7 Classic** by keshikan — **SIL OFL 1.1**, ships TTF + WOFF, 6 styles. Nearest match to the chamfered ends. Alternatives: DSEG7 Modern (squarer), or Google Fonts' **Share Tech Mono** if you want a non-segmented fallback.

8. Documented colour values from modding communities

Honest answer: **the theme-modding scene does not publish hex palettes.** Wii themes are distributed as `.mym` / `.csm` bundles built by **ThemeMii** / **CustomizeMii**, and every visual is a **TPL texture** inside the `layout/` folder of the system menu WAD — creators swap bitmaps and reposition *panes*, they never edit colour constants. So there is no canonical hex list to find; GBAtemp's "Wii Theme Team Creations" thread and the WiiBrew / hacks.guide theme pages document the pipeline and pane structure, not colours.

Therefore treat my sampled palette in §0 as the authoritative source for this build. Caveat: the reference is a slightly desaturated capture (its brightest white is 249, not 255), so if you want a cleaner look, scale every value up by ~2.5 % rather than colour-picking individually — that keeps relationships intact.

9. Motion timing

Transition	Timing	Confidence
Base frame rate for all system animation	**59.94 Hz** (NTSC); PAL50 = 1.2× speed; animations shorter than 1.2 frames may drop	**S** Nintendo spec
Health & Safety warning → menu	Screen holds until A is pressed, then **fades out** (not a cut). Fade ≈ 400–600 ms	**S** behaviour / *est.* duration
Startup chime length	**~1.0–1.3 s**, overlapping the fade-in	**S**
Logo → menu fade-in	≈ 500 ms crossfade; grid tiles settle in, no stagger	*est.*
Channel hover in	≈ 180–220 ms, single spring overshoot	*est.*
Channel hover out	≈ 120–150 ms ease-out, no overshoot	*est.*
Channel launch zoom-in	≈ 400–500 ms tile → full-screen banner; banner sound is gated to *after* the zoom completes	**S** ordering / *est.* duration
"Start" pressed → game	Banner sound stops at **exactly 2 s**	**S** Nintendo spec
"Wii Menu" pressed → return	Banner + sound fade out together, then zoom back to tile	**S**
Icon animation loop	Spec's worked example: frames **0** → **2400** = **40.0 s** at 60 fps	**S**

Sources

Nintendo, *Icon and Banner Specifications* RVL-06-0166-001-L v1.0.0 (2008) · Nintendo, *Wii Operations Manual — Channels and Settings* MAB-RVK-S-USZ-CO · "The Nintendo Wii's UI Sounds", Beyond the Beep · NintendoWiki, *List of Nintendo system fonts* · Fontendo (Wii v0.1 Futura placeholder) · keshikan/DSEG (GitHub, OFL 1.1) · Google Fonts, M PLUS Rounded 1c · docs.oscwii.org (Wii Shop Channel UI sound table) · The Sounds Resource / Internet Archive / KHInsider (audio rips) · WiiBrew & wiki.hacks.guide (Wii Menu Themes, ThemeMii/CustomizeMii pipeline) · GBAtemp "Wii Theme Team Creations".